The empirical recurrence for OEIS A283542: recovering a conjecture that is not written down

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Abstract

OEIS A283542 records "Empirical recurrence of order 59", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 80$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 59, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 59 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283542 is "Number of nX7 0..1 arrays with no 1 equal to more than one of its horizontal, diagonal and antidiagonal neighbors.". It has offset 1 and begins

$$81, 1444, 42664, 1094959, 29377334, 778500603, 20707472573, 550220030803, \ \ldots$$

The entry states:

> Empirical recurrence of order 59 (see link above)

As of the "Last modified" line on the live entry (Mar 10 2017 09:30:13, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1525$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 80$ of the 1525, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 80$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 160$ exact terms of the model returns order 59 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{59} c_i\, a(n-i),$$

c_{1}	33	c_{16}	145181326142	c_{31}	519541275639249	c_{46}	−463878419113O2984
c_{2}	−127	c_{17}	1793059032976	c_{32}	1661569099047225	c_{47}	14285777783172020
c_{3}	−2273	c_{18}	1135845461934	c_{33}	−2099492444405708	c_{48}	23490112378102984
c_{4}	29843	c_{19}	−8185845833946	c_{34}	−2272407955195800	c_{49}	−15020712477177616
c_{5}	30321	c_{20}	−13783961187661	c_{35}	3319367240679904	c_{50}	−5570697670139840
c_{6}	−1162127	c_{21}	32469751627159	c_{36}	5385243402665705	c_{51}	7283836840922624
c_{7}	3408537	c_{22}	36583804163505	c_{37}	−6416298174192869	c_{52}	1240809736103680
c_{8}	20633529	c_{23}	−38264233320917	c_{38}	−4382018703851740	c_{53}	−2604637883058176
c_{9}	−175120761	c_{24}	−149904117293915	c_{39}	2384918557987151	c_{54}	114583021821952
c_{10}	−290981528	c_{25}	128124550994860	c_{40}	343392207656810	c_{55}	457857599406080
c_{11}	2537789513	c_{26}	170076900926407	c_{41}	17681096322161596	c_{56}	−39325947920384
c_{12}	2030089504	c_{27}	−39935503784642	c_{42}	−17532569301642522	c_{57}	−64491031101440
c_{13}	15643364716	c_{28}	−188975757378205	c_{43}	−25368832698922886	c_{58}	2992518463488
c_{14}	−86938218786	c_{29}	−210744587327353	c_{44}	47404417170117671	c_{59}	2576980377600
c_{15}	−163768766369	c_{30}	−222134843658175	c_{45}	2375033240906577		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 59, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $59 + 4$ terms removed returns order 59 as well, so $\deg q_{\min} = 59 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. ☐

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $59+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 59 that this sequence satisfies — and that family turns out to have exactly one member.

References

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